Project Title

Accident Hotspot Prediction Using Random Forest and Spatial Clustering

Project Report submitted for the partial fulfilment of grade for the course

MACHINE LEARNING (25SC2107E)

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Submitted by
G. CHERAN TEJ – 2520030466
A. ARUN TEJ – 2520030472
E. ASHWITH REDDY - 2520030475

Team Number: 16
Section: 6

Under the guidance of

GADDAM RAVINDRA BABU

(DEEMED TO BE UNIVERSITY)
BACHUPALLY

Hyderabad-500090, Telangana, India.

ABSTRACT

Road accidents are a major public safety concern, causing significant loss of life, injuries, and economic damage every year. Identifying accident-prone locations in advance can help authorities implement effective preventive measures. This project presents an **Accident Hotspot Prediction System** using **Random Forest** and **Spatial Clustering** techniques to analyze historical road accident data. The system considers factors such as accident location, time, weather conditions, road type, traffic density, and vehicle information to predict the likelihood of accidents. Random Forest, a robust machine learning algorithm, is used to classify accident risk by learning complex relationships among multiple features and providing accurate predictions.

Spatial Clustering groups geographically close accident locations to identify high-risk zones, enabling visualization of accident hotspots on a map. The combined approach improves prediction accuracy while offering valuable geographical insights for decision-makers. The proposed system assists traffic management authorities, urban planners, and emergency response teams in prioritizing safety measures, optimizing resource allocation, and reducing accident rates. Ultimately, this solution contributes to safer transportation systems by supporting data-driven planning and proactive road safety management.

Signature of the Guide